🗄️ نظام إدارة الفعاليات - Database Schema Documentation

Event Management System - Complete Database Schema

📊 Database Overview - نظرة عامة على قاعدة البيانات

Total Tables: 21 جدول

- **8 جداول أنواع/فئات (Lookup Tables)** - للبيانات المرجعية القابلة للتخصيص
- **12 جدول رئيسي (Main Tables)** - لبيانات النظام الأساسية
- **1 جدول DTO** - للوحة التحكم والتقارير

🎯 Section 1: Lookup/Reference Tables - جداول الأنواع والفئات

هذه الجداول تحتوي على البيانات المرجعية القابلة للتخصيص والإدارة من لوحة التحكم.

1.1 Event Types - أنواع الفعاليات

Table: `event_types`

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع
TypeCode	string(50)	Unique type code	رمز النوع الفريد
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon/emoji	الأيقونة
ColorCode	string(20)	Color code (hex)	رمز اللون
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_types VALUES
('conference', 'Conference', 'مؤتمر', 'Professional conferences', '🎤',
```

```
'#3498db', 1, true),
('exhibition', 'Exhibition', 'معرض', 'Trade exhibitions', '🏪', '#2ecc71', 2,
true),
('festival', 'Festival', 'مهرجان', 'Festivals and celebrations', '🎉',
'#e74c3c', 3, true),
('concert', 'Concert', 'حفل موسيقي', 'Musical concerts', '🎵', '#9b59b6', 4,
true),
('sports', 'Sports Event', 'حدث رياضي', 'Sports events', '⚽', '#f39c12', 5,
true),
('workshop', 'Workshop', 'ورشة عمل', 'Training workshops', '🛠️', '#1abc9c',
6, true);
```

1.2 Zone Types - أنواع المناطق

Table: `event_zone_types`

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع
TypeCode	string(50)	Unique type code	رمز النوع
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
RequiresSpecialAccess	bool	Requires special access	يتطلب صلاحية خاصة
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_zone_types VALUES
('vip', 'VIP Area', 'منطقة VIP', 'VIP exclusive area', '👑', '#FFD700', true,
1, true),
('general', 'General Area', 'منطقة عامة', 'Public general area', '👥',
'#3498db', false, 2, true),
('food', 'Food & Beverage', 'منطقة الطعام', 'Food court area', '🍽️',
'#e67e22', false, 3, true),
('exhibition', 'Exhibition Area', 'منطقة المعارض', 'Exhibition halls', '🏛️',
'#9b59b6', false, 4, true),
('parking', 'Parking Area', 'منطقة الانتظار', 'Vehicle parking', '🚗',
'#95a5a6', false, 5, true),
```

```
('stage', 'Main Stage', 'المسرح الرئيسي', 'Main event stage', '🎭',
'#e74c3c', true, 6, true);
```

1.3 Gate Types - أنواع البوابات

Table: `event_gate_types`

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع
TypeCode	string(50)	Unique type code	رمز النوع
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
Direction	string(20)	Direction type	نوع الاتجاه
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_gate_types VALUES
('entry', 'Entry Gate', 'بوابة دخول', 'Entry only gate', '🚪➡️', '#2ecc71',
'entry', 1, true),
('exit', 'Exit Gate', 'بوابة خروج', 'Exit only gate', '🚪⬅️', '#e74c3c',
'exit', 2, true),
('both', 'Two-Way Gate', 'بوابة ثنائية', 'Entry and exit', '🚪↔️', '#3498db',
'both', 3, true),
('vip', 'VIP Gate', 'بوابة VIP', 'VIP exclusive', '👑🚪', '#FFD700', 'both',
4, true),
('staff', 'Staff Gate', 'بوابة الموظفين', 'Staff only', '👔🚪', '#95a5a6',
'both', 5, true),
('emergency', 'Emergency Exit', 'مخرج طوارئ', 'Emergency use only', '🚨🚪',
'#c0392b', 'exit', 6, true);
```

1.4 Visitor Types - أنواع الزوار

Table: `event_visitor_types`

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع
TypeCode	string(50)	Unique type code	رمز النوع
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
BadgeColor	string(20)	Badge color	لون البطاقة
HasPriorityAccess	bool	Priority access	أولوية الدخول
RequiresApproval	bool	Requires approval	يتطلب موافقة
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_visitor_types VALUES
('regular', 'Regular Visitor', 'زائر عادي', 'Standard visitor', '👤',
'#3498db', '#FFFFFF', false, false, 1, true),
('vip', 'VIP Guest', 'ضيف VIP', 'VIP guest', '⭐', '#FFD700', '#FFD700', true,
false, 2, true),
('speaker', 'Speaker', 'متحدث', 'Event speaker', '🎤', '#e74c3c', '#FF6B6B',
true, true, 3, true),
('media', 'Media/Press', 'إعلامي', 'Media representative', '📰', '#9b59b6',
'#CE93D8', true, true, 4, true),
('exhibitor', 'Exhibitor', 'عارض', 'Booth exhibitor', '🏪', '#f39c12',
'#FFB74D', true, false, 5, true),
('sponsor', 'Sponsor', 'راعي', 'Event sponsor', '💎', '#1abc9c', '#4DB6AC',
true, true, 6, true),
('student', 'Student', 'طالب', 'Student visitor', '🎓', '#16a085', '#81C784',
false, false, 7, true);
```

1.5 Ticket Types - أنواع التذاكر

Table: `event_ticket_types`

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع

Column	Type	Description (EN)	الوصف (AR)
TypeCode	string(50)	Unique type code	رمز النوع
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
DefaultPrice	decimal	Default price	السعر الافتراضي
Currency	string(10)	Currency	العملة
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
IsTransferable	bool	Can be transferred	قابل للتحويل
IsRefundable	bool	Can be refunded	قابل للاسترجاع
MaxQuantityPerPerson	int	Max per person	الحد الأقصى للشخص
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_ticket_types VALUES
('free', 'Free Entry', 'دخول مجاني', 'Free admission', 0.00, 'SAR', '🎫',
'#2ecc71', false, false, 1, 1, true),
('general', 'General Admission', 'دخول عام', 'Standard ticket', 100.00, 'SAR',
'🎟️', '#3498db', true, true, 5, 2, true),
('vip', 'VIP Ticket', 'تذكرة VIP', 'VIP access', 500.00, 'SAR', '⭐',
'#FFD700', false, false, 2, 3, true),
('premium', 'Premium Pass', 'تذكرة بريميوم', 'Premium features', 300.00,
'SAR', '💎', '#9b59b6', true, true, 3, 4, true),
('early_bird', 'Early Bird', 'الحجز المبكر', 'Discount price', 80.00, 'SAR',
'🐦', '#f39c12', true, true, 10, 5, true),
('group', 'Group Ticket', 'تذكرة مجموعة', 'Group discount', 400.00, 'SAR',
'👥', '#1abc9c', true, true, 10, 6, true);
```

1.6 Permit Types - أنواع التصاريح

Table: `event_permit_types`

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع

Column	Type	Description (EN)	الوصف (AR)
TypeCode	string(50)	Unique type code	رمز النوع
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
RequiresApproval	bool	Needs approval	يتطلب موافقة
ApprovalLevel	string(50)	Approval level	مستوى الموافقة
DefaultValidityDays	int	Default validity	أيام الصلاحية
RequiresVehicleInfo	bool	Vehicle info needed	يتطلب معلومات مركبة
RequiresPhoto	bool	Photo required	يتطلب صورة
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_permit_types VALUES
('staff', 'Staff Permit', 'تصريح موظف', 'Event staff', '🦺', '#3498db', true,
'supervisor', 30, false, true, 1, true),
('vendor', 'Vendor Permit', 'تصريح مورد', 'Service vendor', '🛠️', '#f39c12',
true, 'manager', 7, false, true, 2, true),
('contractor', 'Contractor Pass', 'تصريح مقاول', 'Construction contractor',
'👷', '#e67e22', true, 'manager', 15, false, true, 3, true),
('vehicle', 'Vehicle Permit', 'تصريح مركبة', 'Vehicle access', '🚗',
'#95a5a6', true, 'security', 7, true, false, 4, true),
('media', 'Media Pass', 'تصريح إعلامي', 'Press/media', '📰', '#9b59b6', true,
'pr_manager', 3, false, true, 5, true),
('vip', 'VIP Access', 'تصريح VIP', 'VIP backstage', '⭐', '#FFD700', true,
'director', 30, false, true, 6, true),
('emergency', 'Emergency Services', 'خدمات طوارئ', 'Emergency personnel',
'🚑', '#e74c3c', false, 'auto', 90, true, true, 7, true);
```

1.7 Booth Types - أنواع الأكشاك

Table: `event_booth_types`

Column	Type	Description (EN)	الوصف (AR)

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع
TypeCode	string(50)	Unique type code	رمز النوع
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
DefaultPricePerSqm	decimal	Price per sqm	السعر للمتر المربع
MinSizeSqm	decimal	Min size	الحد الأدنى للمساحة
MaxSizeSqm	decimal	Max size	الحد الأقصى للمساحة
IncludesElectricity	bool	Has electricity	يشمل كهرباء
IncludesInternet	bool	Has internet	يشمل إنترنت
IncludesFurniture	bool	Has furniture	يشمل أثاث
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_booth_types VALUES
('standard', 'Standard Booth', 'كشك قياسي', 'Basic booth', '🏪', '#3498db',
500.00, 9.0, 20.0, true, false, false, 1, true),
('premium', 'Premium Booth', 'كشك بريميوم', 'Premium features', '⭐',
'#FFD700', 800.00, 15.0, 40.0, true, true, true, 2, true),
('food', 'Food Stall', 'كشك طعام', 'Food service', '🍽', '#e67e22', 600.00,
12.0, 30.0, true, false, true, 3, true),
('retail', 'Retail Shop', 'محل بيع', 'Retail store', '🛍', '#9b59b6', 700.00,
15.0, 50.0, true, true, true, 4, true),
('exhibition', 'Exhibition Stand', 'منصة عرض', 'Display stand', '🖼',
'#2ecc71', 1000.00, 20.0, 100.0, true, true, false, 5, true),
('kiosk', 'Small Kiosk', 'كشك صغير', 'Compact kiosk', '📦', '#1abc9c',
300.00, 4.0, 9.0, true, false, false, 6, true);
```

1.8 Incident Types - أنواع الحوادث

Table: `event_incident_types`

Column	Type	Description (EN)	الوصف (AR)
TypeId	long (PK)	Type ID	معرف النوع
TypeCode	string(50)	Unique type code	رمز النوع
TypeName	string(100)	Type name (English)	اسم النوع بالإنجليزية
TypeNameAr	string(100)	Type name (Arabic)	اسم النوع بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
DefaultPriority	string(20)	Default priority	الأولوية الافتراضية
DefaultSeverity	string(20)	Default severity	الخطورة الافتراضية
AssignedTeam	string(100)	Responsible team	الفريق المسؤول
SLAResponseMinutes	int	Response SLA	وقت الاستجابة
SLAResolutionMinutes	int	Resolution SLA	وقت الحل
RequiresImmediateAction	bool	Immediate action	إجراء فوري
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_incident_types VALUES
('security', 'Security Issue', 'مشكلة أمنية', 'Security incident', '🔒',
'#e74c3c', 'high', 'major', 'Security Team', 5, 30, true, 1, true),
('medical', 'Medical Emergency', 'طوارئ طبية', 'Medical issue', '🚑',
'#c0392b', 'critical', 'critical', 'Medical Team', 2, 10, true, 2, true),
('technical', 'Technical Issue', 'مشكلة تقنية', 'Tech problem', '⚙️',
'#3498db', 'medium', 'moderate', 'IT Support', 15, 60, false, 3, true),
('maintenance', 'Maintenance Request', 'طلب صيانة', 'Maintenance needed',
'🔧', '#f39c12', 'low', 'minor', 'Maintenance', 30, 120, false, 4, true),
('complaint', 'Visitor Complaint', 'شكوى زائر', 'Customer complaint', '📝',
'#95a5a6', 'medium', 'moderate', 'Customer Service', 10, 120, false, 5, true),
('fire', 'Fire Hazard', 'خطر حريق', 'Fire emergency', '🔥', '#e74c3c',
'critical', 'critical', 'Fire Safety', 1, 5, true, 6, true),
('crowd', 'Crowd Control', 'إدارة الحشود', 'Crowd issue', '👥', '#9b59b6',
'high', 'major', 'Security Team', 5, 15, true, 7, true),
('lost_found', 'Lost & Found', 'مفقودات', 'Lost item', '🔍', '#1abc9c',
'low', 'minor', 'Information Desk', 30, 180, false, 8, true);
```

1.9 Payment Methods - طرق الدفع

Table: `event_payment_methods`

Column	Type	Description (EN)	الوصف (AR)
MethodId	long (PK)	Method ID	معرف الطريقة
MethodCode	string(50)	Unique method code	رمز الطريقة
MethodName	string(100)	Method name (English)	اسم الطريقة بالإنجليزية
MethodNameAr	string(100)	Method name (Arabic)	اسم الطريقة بالعربية
Description	string(500)	Description	الوصف
Icon	string(100)	Icon	الأيقونة
ColorCode	string(20)	Color code	رمز اللون
PaymentGateway	string(50)	Gateway name	اسم البوابة
GatewayConfig	string(2000)	Gateway config (JSON)	إعدادات البوابة
TransactionFeePercentage	decimal	Fee percentage	نسبة الرسوم
FixedTransactionFee	decimal	Fixed fee	رسوم ثابتة
IsOnlinePayment	bool	Is online	دفع أونلاين
RequiresVerification	bool	Needs verification	يتطلب تحقق
DisplayOrder	int	Display order	ترتيب العرض
IsActive	bool	Active status	حالة التفعيل

Sample Data / بيانات نموذجية:

```
INSERT INTO event_payment_methods VALUES
('cash', 'Cash', 'نقدي', 'Cash payment', '💵', '#2ecc71', null, null, 0.00,
0.00, false, false, 1, true),
('card', 'Credit/Debit Card', 'بطاقة', 'Card payment', '💳', '#3498db',
'Paymob', '{}', 2.75, 0.00, true, true, 2, true),
('mada', 'Mada', 'مدى', 'Mada card', '💳', '#00A651', 'Paymob', '{}', 2.00,
0.00, true, true, 3, true),
('bank_transfer', 'Bank Transfer', 'تحويل بنكي', 'Bank transfer', '🏦',
'#9b59b6', null, null, 0.00, 5.00, false, true, 4, true),
('apple_pay', 'Apple Pay', 'آبل باي', 'Apple Pay', '', '#000000', 'Paymob',
'{}', 2.75, 0.00, true, true, 5, true),
('stc_pay', 'STC Pay', 'اس تي سي باي', 'STC mobile payment', '📱', '#6e1a5d',
'STC', '{}', 2.50, 0.00, true, true, 6, true);
```

📋 Section 2: Main Entity Tables - الجداول الرئيسية

2.1 Events - الفعاليات

Table: `event_events`

Purpose: Store main event information / تخزين معلومات الفعاليات الرئيسية

Column	Type	Nullable	Description
EventId	long (PK)	No	Unique event ID
EventCode	string(50)	No	Event code
EventName	string(200)	No	Event name (English)
EventNameAr	string(200)	Yes	Event name (Arabic)
Description	string(2000)	Yes	Description (English)
DescriptionAr	string(2000)	Yes	Description (Arabic)
EventType	string(50)	Yes	Event type (from event_types)
Status	string(20)	No	Status (planned, active, completed, cancelled)
StartDate	DateTime	No	Start date
EndDate	DateTime	No	End date
RegistrationStartDate	DateTime	Yes	Registration start
RegistrationEndDate	DateTime	Yes	Registration end
LocationId	long	Yes	Location reference
LocationName	string(200)	Yes	Location name
Address	string(500)	Yes	Full address
City	string(100)	Yes	City
Latitude	decimal	Yes	GPS latitude
Longitude	decimal	Yes	GPS longitude
MapImageUrl	string(500)	Yes	Map image URL
ExpectedVisitors	int	Yes	Expected visitor count
MaxCapacity	int	Yes	Maximum capacity
CurrentOccupancy	int	No	Current occupancy (computed)
Organizer	string(200)	Yes	Organizer name
ContactEmail	string(100)	Yes	Contact email
ContactPhone	string(20)	Yes	Contact phone
Website	string(200)	Yes	Website URL
LogoUrl	string(500)	Yes	Logo URL

Column	Type	Nullable	Description
BannerImageUrl	string(500)	Yes	Banner image URL
IsPublic	bool	No	Public event
RequiresRegistration	bool	No	Registration required
IsPaidEvent	bool	No	Paid event
TicketPrice	decimal	Yes	Ticket price
Currency	string(10)	Yes	Currency code
EventManagerId	long	Yes	Event manager ID
Tags	string(500)	Yes	Tags (comma-separated)
IsActive	bool	No	Active status
+ SysBase fields	-	-	Audit fields

2.2 Event Zones - المناطق

Table: `event_zones`

Purpose: Define zones/areas within an event / تحديد المناطق داخل الفعالية

Column	Type	Description
ZoneId	long (PK)	Zone ID
EventId	long (FK)	Event reference
ZoneCode	string(50)	Zone code
ZoneName	string(200)	Zone name (English)
ZoneNameAr	string(200)	Zone name (Arabic)
Description	string(1000)	Description
ZoneType	string(50)	Zone type (from zone_types)
ParentZoneId	long	Parent zone (for sub-zones)
MaxCapacity	int	Maximum capacity
CurrentOccupancy	int	Current occupancy
AreaSqm	decimal	Area in square meters
FloorNumber	int	Floor number
BuildingName	string(100)	Building name
MapCoordinates	string(2000)	Map coordinates (JSON)

Column	Type	Description
ColorCode	string(20)	Display color
AccessLevel	string(20)	Access level (public, restricted, vip)
ZoneManagerId	long	Zone manager ID
Status	string(20)	Status
DisplayOrder	int	Display order
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.3 Event Gates - البوابات

Table: `event_gates`

Purpose: Manage entry/exit gates / إدارة بوابات الدخول والخروج

Column	Type	Description
GateId	long (PK)	Gate ID
EventId	long (FK)	Event reference
ZoneId	long (FK)	Zone reference (optional)
GateCode	string(50)	Gate code
GateName	string(200)	Gate name (English)
GateNameAr	string(200)	Gate name (Arabic)
Description	string(1000)	Description
GateType	string(20)	Gate type (from gate_types)
Location	string(200)	Location description
Latitude	decimal	GPS latitude
Longitude	decimal	GPS longitude
MaxCapacityPerHour	int	Hourly capacity
CurrentQueueCount	int	Current queue
AvgWaitTimeMinutes	int	Average wait time
Status	string(20)	Status (open, closed, maintenance, full)
OpeningTime	TimeSpan	Opening time
ClosingTime	TimeSpan	Closing time

Column	Type	Description
HasScanner	bool	Has QR scanner
ScannerDeviceId	string(100)	Scanner device ID
GateManagerId	long	Gate manager ID
DisplayOrder	int	Display order
IsActive	bool	Active status
LastStatusUpdate	DateTime	Last status update
+ SysBase fields	-	Audit fields

2.4 Event Visitors - الزوار

Table: `event_visitors`

Purpose: Store visitor information / تخزين معلومات الزوار

Column	Type	Description
VisitorId	long (PK)	Visitor ID
EventId	long (FK)	Event reference
UserId	long	User account ID (optional)
VisitorCode	string(50)	Visitor code
FirstName	string(100)	First name
LastName	string(100)	Last name
FullNameAr	string(200)	Full name (Arabic)
Email	string(100)	Email
Phone	string(20)	Phone
NationalId	string(50)	National ID / Iqama
PassportNumber	string(50)	Passport number
Nationality	string(50)	Nationality
Gender	string(10)	Gender
DateOfBirth	DateTime	Date of birth
CompanyName	string(200)	Company name
JobTitle	string(100)	Job title
VisitorType	string(20)	Visitor type (from visitor_types)

Column	Type	Description
RegistrationDate	DateTime	Registration date
RegistrationSource	string(20)	Registration source
QRCode	string(200)	QR code
BadgeNumber	string(50)	Badge number
BadgePrinted	bool	Badge printed
BadgePrintedDate	DateTime	Badge print date
ProfilePictureUrl	string(500)	Profile picture URL
Status	string(20)	Status (registered, checked-in, checked-out, cancelled)
CheckInTime	DateTime	Check-in time
CheckOutTime	DateTime	Check-out time
CheckInGateId	long	Check-in gate ID
EmergencyContactName	string(100)	Emergency contact
EmergencyContactPhone	string(20)	Emergency phone
SpecialNeeds	string(500)	Special needs
Notes	string(1000)	Notes
IsVIP	bool	VIP status
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.5 Event Tickets - التذاكر

Table: `event_tickets`

Purpose: Manage event tickets / إدارة تذاكر الفعاليات

Column	Type	Description
TicketId	long (PK)	Ticket ID
EventId	long (FK)	Event reference
VisitorId	long (FK)	Visitor reference
TicketNumber	string(50)	Ticket number
TicketType	string(50)	Ticket type (from ticket_types)
TicketCategory	string(100)	Ticket category

Column	Type	Description
Price	decimal	Price
Currency	string(10)	Currency
PurchaseDate	DateTime	Purchase date
ValidFrom	DateTime	Valid from
ValidUntil	DateTime	Valid until
QRCode	string(200)	QR code
Barcode	string(100)	Barcode
Status	string(20)	Status (active, used, cancelled, expired)
IsPaid	bool	Payment status
PaymentMethod	string(50)	Payment method
PaymentReference	string(100)	Payment reference
UsedDate	DateTime	Used date
UsedGateId	long	Gate used at
SeatNumber	string(50)	Seat number
ZoneAccess	string(500)	Zone access (JSON)
CanBeTransferred	bool	Transferable
TransferredToVisitorId	long	Transferred to
TransferDate	DateTime	Transfer date
Notes	string(1000)	Notes
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.6 Event Permits - التصاريح

Table: `event_permits`

Purpose: Manage access permits / إدارة تصاريح الدخول

Column	Type	Description
PermitId	long (PK)	Permit ID
EventId	long (FK)	Event reference
EmployeeId	long	Employee reference

Column	Type	Description
VisitorId	long	Visitor reference
PermitNumber	string(50)	Permit number
PermitType	string(50)	Permit type (from permit_types)
PermitCategory	string(100)	Permit category
ApplicantName	string(200)	Applicant name (English)
ApplicantNameAr	string(200)	Applicant name (Arabic)
CompanyName	string(200)	Company name
NationalId	string(50)	National ID
ContactPhone	string(20)	Contact phone
Email	string(100)	Email
ValidFrom	DateTime	Valid from
ValidUntil	DateTime	Valid until
IssueDate	DateTime	Issue date
IssuedBy	string(100)	Issued by
QRCode	string(200)	QR code
ZoneAccess	string(1000)	Zone access (JSON)
GateAccess	string(1000)	Gate access (JSON)
AccessLevel	string(20)	Access level
Purpose	string(500)	Purpose
VehiclePlateNumber	string(20)	Vehicle plate
VehicleType	string(50)	Vehicle type
PhotoUrl	string(500)	Photo URL
AttachmentUrl	string(500)	Attachment URL
Status	string(20)	Status (pending, approved, rejected, active, expired, cancelled)
ApprovedBy	string(100)	Approved by
ApprovalDate	DateTime	Approval date
RejectionReason	string(500)	Rejection reason
Notes	string(1000)	Notes
IsActive	bool	Active status

Column	Type	Description
+ SysBase fields	-	Audit fields

2.7 Event Booths - الأكشاك

Table: `event_booths`

Purpose: Manage booth/stall rentals / إدارة الأكشاك المؤجرة

Column	Type	Description
BoothId	long (PK)	Booth ID
EventId	long (FK)	Event reference
ZoneId	long (FK)	Zone reference
BoothNumber	string(50)	Booth number
BoothName	string(200)	Booth name (English)
BoothNameAr	string(200)	Booth name (Arabic)
Description	string(1000)	Description
BoothType	string(50)	Booth type (from booth_types)
BoothCategory	string(100)	Booth category
SizeSqm	decimal	Size in sqm
LocationDescription	string(200)	Location
FloorNumber	int	Floor number
MapCoordinates	string(500)	Map coordinates
RentalPrice	decimal	Rental price
Currency	string(10)	Currency
RentalPeriodType	string(20)	Period type
TenantId	long	Tenant ID
ContractId	long	Contract ID
RentalStatus	string(20)	Status (available, rented, reserved, maintenance)
HasElectricity	bool	Has electricity
HasWater	bool	Has water
HasInternet	bool	Has internet
FurnitureIncluded	bool	Includes furniture

Column	Type	Description
FurnitureDescription	string(500)	Furniture details
AdditionalAmenities	string(1000)	Additional amenities
SetupDate	DateTime	Setup date
TeardownDate	DateTime	Teardown date
FloorPlanImageUrl	string(500)	Floor plan image
Notes	string(1000)	Notes
DisplayOrder	int	Display order
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.8 Event Tenants - المستأجرون

Table: `event_tenants`

Purpose: Manage booth tenants / إدارة مستأجري الأكشاك

Column	Type	Description
TenantId	long (PK)	Tenant ID
EventId	long (FK)	Event reference
TenantCode	string(50)	Tenant code
CompanyName	string(200)	Company name (English)
CompanyNameAr	string(200)	Company name (Arabic)
CommercialRegNumber	string(50)	Commercial registration
TaxNumber	string(50)	Tax number
BusinessType	string(100)	Business type
ContactPersonName	string(200)	Contact person
ContactPosition	string(100)	Contact position
Email	string(100)	Email
Phone	string(20)	Phone
Mobile	string(20)	Mobile
Address	string(500)	Address
City	string(100)	City

Column	Type	Description
Country	string(100)	Country
Website	string(200)	Website
LogoUrl	string(500)	Logo URL
BankName	string(100)	Bank name
AccountNumber	string(50)	Account number
IBAN	string(50)	IBAN
RegistrationDate	DateTime	Registration date
Status	string(20)	Status (active, inactive, blacklisted, pending)
Rating	decimal	Rating
TotalBoothsRented	int	Total booths (computed)
Notes	string(1000)	Notes
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.9 Event Contracts - العقود

Table: `event_contracts`

Purpose: Manage rental contracts / إدارة عقود الإيجار

Column	Type	Description
ContractId	long (PK)	Contract ID
EventId	long (FK)	Event reference
TenantId	long (FK)	Tenant reference
BoothId	long (FK)	Booth reference
ContractNumber	string(50)	Contract number
ContractDate	DateTime	Contract date
StartDate	DateTime	Start date
EndDate	DateTime	End date
TotalAmount	decimal	Total amount
Currency	string(10)	Currency
VATAmount	decimal	VAT amount

Column	Type	Description
VATPercentage	decimal	VAT percentage
DepositAmount	decimal	Deposit amount
PaymentTerms	string(500)	Payment terms
PaymentMethod	string(50)	Payment method
PaymentSchedule	string(2000)	Payment schedule (JSON)
Status	string(20)	Status (draft, pending, active, completed, cancelled, terminated)
SignedDate	DateTime	Signed date
SignedByTenant	bool	Tenant signature
SignedByOrganizer	bool	Organizer signature
ContractFileUrl	string(500)	Contract file URL
TermsAndConditions	string(4000)	Terms and conditions
SpecialConditions	string(2000)	Special conditions
AutoRenewal	bool	Auto renewal
RenewalDate	DateTime	Renewal date
CancellationReason	string(500)	Cancellation reason
CancelledBy	string(100)	Cancelled by
CancellationDate	DateTime	Cancellation date
Notes	string(1000)	Notes
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.10 Event Incidents - الحوادث

Table: `event_incidents`

Purpose: Manage incidents and support tickets / إدارة الحوادث والبلاغات

Column	Type	Description
IncidentId	long (PK)	Incident ID
EventId	long (FK)	Event reference
IncidentNumber	string(50)	Incident number
IncidentType	string(50)	Incident type (from incident_types)

Column	Type	Description
IncidentCategory	string(100)	Incident category
Priority	string(20)	Priority (low, medium, high, critical)
Severity	string(20)	Severity (minor, moderate, major, critical)
Title	string(200)	Title
Description	string(2000)	Description
Location	string(200)	Location
ZoneId	long	Zone reference
BoothId	long	Booth reference
ReportedDate	DateTime	Reported date
ReportedBy	string(100)	Reported by
ReporterType	string(50)	Reporter type
ReporterContact	string(100)	Reporter contact
Status	string(20)	Status (reported, assigned, in-progress, resolved, closed, escalated)
AssignedToId	long	Assigned to
AssignedDate	DateTime	Assigned date
AssignedTeam	string(100)	Assigned team
ResponseTimeMinutes	int	Response time
ResolutionTimeMinutes	int	Resolution time
ResolutionDate	DateTime	Resolution date
ResolutionNotes	string(2000)	Resolution notes
Escalated	bool	Escalated
EscalatedTo	string(100)	Escalated to
EscalationDate	DateTime	Escalation date
EscalationReason	string(500)	Escalation reason
Attachments	string(2000)	Attachments (JSON)
WitnessInformation	string(1000)	Witness info
FollowUpRequired	bool	Follow-up required
FollowUpDate	DateTime	Follow-up date

Column	Type	Description
ClosedDate	DateTime	Closed date
ClosedBy	string(100)	Closed by
SatisfactionRating	int	Customer satisfaction (1-5)
Notes	string(2000)	Notes
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.11 Event Payments - الدفعات

Table: `event_payments`

Purpose: Track all payments / تتبع جميع الدفعات

Column	Type	Description
PaymentId	long (PK)	Payment ID
EventId	long (FK)	Event reference
ContractId	long	Contract reference
InvoiceId	long	Invoice reference
TenantId	long	Tenant reference
VisitorId	long	Visitor reference
PaymentNumber	string(50)	Payment number
PaymentType	string(50)	Payment type
PaymentDate	DateTime	Payment date
Amount	decimal	Amount
Currency	string(10)	Currency
VATAmount	decimal	VAT amount
TotalAmount	decimal	Total amount
PaymentMethod	string(50)	Payment method (from payment_methods)
CardType	string(50)	Card type
CardLast4	string(4)	Last 4 digits
TransactionReference	string(100)	Transaction reference
BankReference	string(100)	Bank reference

Column	Type	Description
PaymentGateway	string(50)	Payment gateway
GatewayResponse	string(2000)	Gateway response
Status	string(20)	Status (pending, completed, failed, refunded, cancelled)
PayerName	string(200)	Payer name
PayerEmail	string(100)	Payer email
PayerPhone	string(20)	Payer phone
ReceiptNumber	string(50)	Receipt number
ReceiptUrl	string(500)	Receipt URL
RefundAmount	decimal	Refund amount
RefundDate	DateTime	Refund date
RefundReason	string(500)	Refund reason
ProcessedBy	string(100)	Processed by
Notes	string(1000)	Notes
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

2.12 Event Invoices - الفواتير

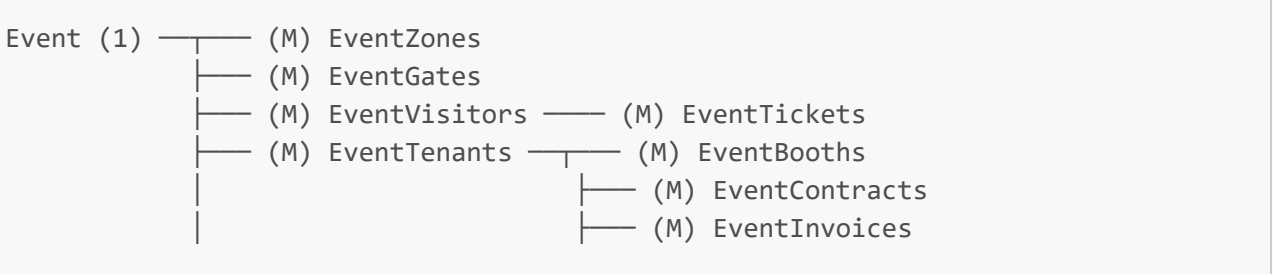
Table: `event_invoices`

Purpose: Manage invoices / إدارة الفواتير

Column	Type	Description
InvoiceId	long (PK)	Invoice ID
EventId	long (FK)	Event reference
ContractId	long	Contract reference
TenantId	long	Tenant reference
InvoiceNumber	string(50)	Invoice number
InvoiceDate	DateTime	Invoice date
DueDate	DateTime	Due date
InvoiceType	string(50)	Invoice type
Description	string(500)	Description

Column	Type	Description
SubtotalAmount	decimal	Subtotal
VATPercentage	decimal	VAT percentage
VATAmount	decimal	VAT amount
DiscountAmount	decimal	Discount
TotalAmount	decimal	Total amount
PaidAmount	decimal	Paid amount
Balance	decimal	Balance
Currency	string(10)	Currency
Status	string(20)	Status (draft, issued, partially_paid, paid, overdue, cancelled)
BillingAddress	string(500)	Billing address
InvoiceItems	string(4000)	Invoice items (JSON)
PaymentTerms	string(500)	Payment terms
InvoiceFileUrl	string(500)	Invoice file URL
SentDate	DateTime	Sent date
SentToEmail	string(100)	Sent to email
PaymentDate	DateTime	Payment date
LateFeeAmount	decimal	Late fee
DaysOverdue	int	Days overdue (computed)
CancelledDate	DateTime	Cancelled date
CancellationReason	string(500)	Cancellation reason
Notes	string(1000)	Notes
IsActive	bool	Active status
+ SysBase fields	-	Audit fields

🔗 Entity Relationships - العلاقات بين الجداول



```
            │                          └── (M) EventPayments
            ├── (M) EventIncidents
            └── (M) EventPermits

EventZone (1) ──┬── (M) EventBooths
                └── (M) EventGates

EventTypes ─── (M) Events
ZoneTypes ─── (M) EventZones
GateTypes ─── (M) EventGates
VisitorTypes ─── (M) EventVisitors
TicketTypes ─── (M) EventTickets
PermitTypes ─── (M) EventPermits
BoothTypes ─── (M) EventBooths
IncidentTypes ─── (M) EventIncidents
PaymentMethods ─── (M) EventPayments
```

Section 3: Database Initialization

تهيئة قاعدة البيانات

لإنشاء جميع الجداول تلقائياً، أضف الكود التالي في `Program.cs`:

```
using ZR.Model.EventManagement;

// Initialize Event Management System tables
app.Environment.InitDbWithTypes(
    // Main Tables
    typeof(Event),
    typeof(EventZone),
    typeof(EventGate),
    typeof(EventVisitor),
    typeof(EventTicket),
    typeof(EventPermit),
    typeof(EventBooth),
    typeof(EventTenant),
    typeof(EventContract),
    typeof(EventIncident),
    typeof(EventPayment),
    typeof(EventInvoice),

    // Lookup Tables
    typeof(EventType),
    typeof(ZoneType),
    typeof(GateType),
    typeof(VisitorType),
    typeof(TicketType),
    typeof(PermitType),
    typeof(BoothType),
```

```
    typeof(IncidentType),
    typeof(PaymentMethod)
);
```

🎯 Section 4: Sample Data Scripts

نصوص البيانات النموذجية

راجع القسم الأول من هذا الملف لجميع نصوص INSERT لتعبئة جداول الأنواع.

🔒 Section 5: Indexes & Performance

الفهارس الموصى بها

```
-- Events
CREATE INDEX idx_event_status ON event_events(Status);
CREATE INDEX idx_event_dates ON event_events(StartDate, EndDate);
CREATE INDEX idx_event_type ON event_events(EventType);

-- Visitors
CREATE INDEX idx_visitor_event ON event_visitors(EventId);
CREATE INDEX idx_visitor_status ON event_visitors(Status);
CREATE INDEX idx_visitor_qr ON event_visitors(QRCode);
CREATE INDEX idx_visitor_email ON event_visitors(Email);

-- Gates
CREATE INDEX idx_gate_event ON event_gates(EventId);
CREATE INDEX idx_gate_status ON event_gates(Status);

-- Incidents
CREATE INDEX idx_incident_event ON event_incidents(EventId);
CREATE INDEX idx_incident_status ON event_incidents(Status);
CREATE INDEX idx_incident_priority ON event_incidents(Priority);

-- Payments
CREATE INDEX idx_payment_event ON event_payments(EventId);
CREATE INDEX idx_payment_tenant ON event_payments(TenantId);
CREATE INDEX idx_payment_status ON event_payments(Status);

-- Invoices
CREATE INDEX idx_invoice_tenant ON event_invoices(TenantId);
CREATE INDEX idx_invoice_status ON event_invoices(Status);
CREATE INDEX idx_invoice_due ON event_invoices(DueDate);
```

📊 Section 6: Database Statistics

إحصائيات قاعدة البيانات

Category	Count
Total Tables	21
Lookup Tables	8
Main Entity Tables	12
Dashboard DTOs	1
Total Columns	~600+
Foreign Keys	~30
Indexes (Recommended)	~20

✨ Key Features - المزايا الرئيسية

1. ☑ **Fully Normalized** - قاعدة بيانات منظمة بالكامل
2. ☑ **Bilingual Support** - دعم العربية والإنجليزية
3. ☑ **Flexible Type System** - نظام أنواع مرن وقابل للتخصيص
4. ☑ **Audit Trail** - تتبع كامل للتعديلات
5. ☑ **Multi-tenancy Ready** - جاهز لدعم متعدد المستأجرين
6. ☑ **Scalable Design** - تصميم قابل للتوسع
7. ☑ **Rich Metadata** - بيانات وصفية غنية
8. ☑ **Performance Optimized** - محسّن للأداء

📞 Support

للمزيد من المعلومات:

- **EVENT_MANAGEMENT_SYSTEM.md** - التوثيق الشامل
- **EVENT_MANAGEMENT_QUICKSTART.md** - دليل البداية السريعة

✨ تم التوثيق بحمد الله
Database Schema Documentation Complete 🎉